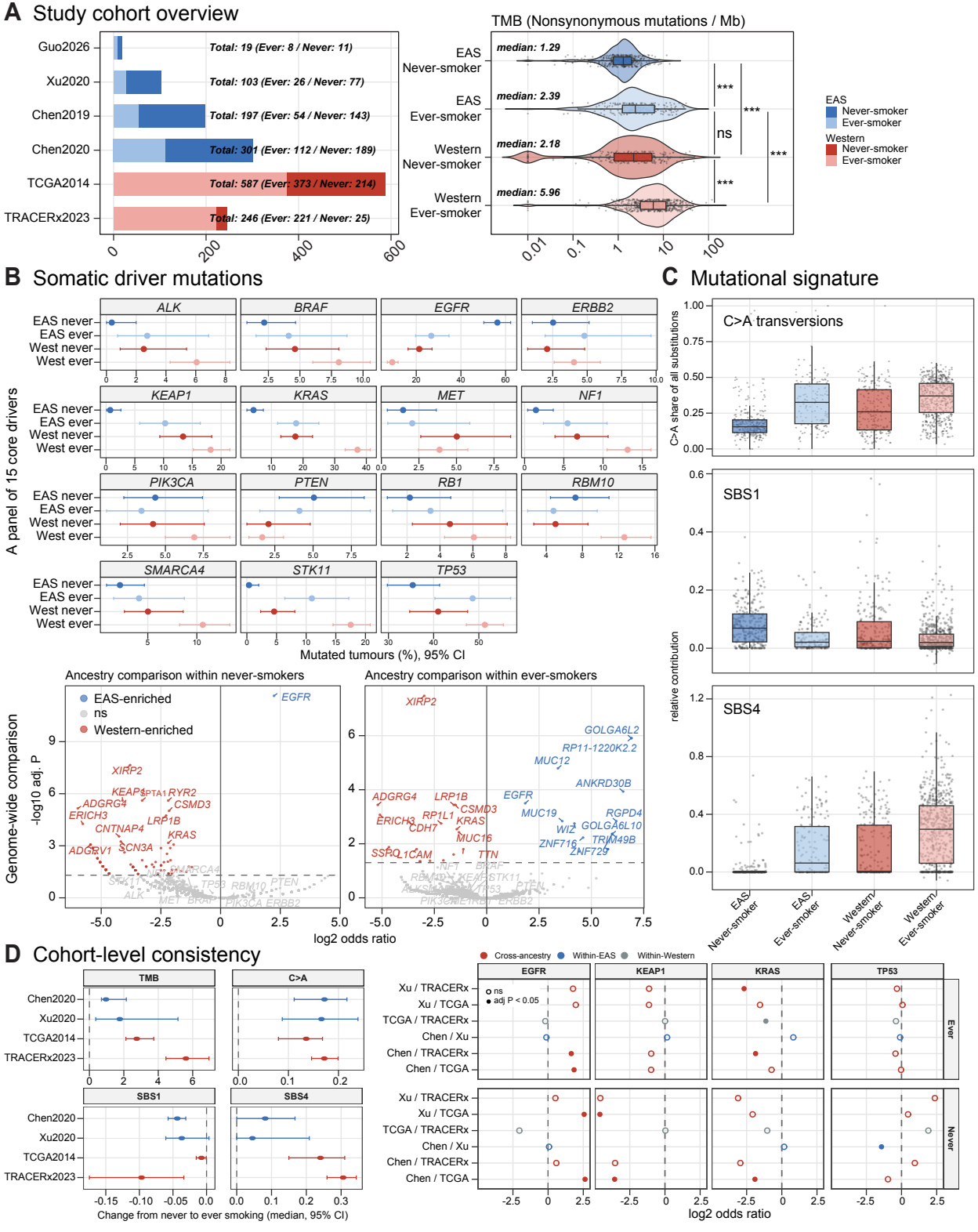

A Study cohort overview
Guo2026
Xu2020
Chen2019
Chen2020
TCGA2014
TRACERx2023
Total: 19 (Ever: 8 / Never: 11)
Total: 103 (Ever: 26 / Never: 77)
Total: 197 (Ever: 54 / Never: 143)
Total: 301 (Ever: 112 / Never: 189)
Total: 587 (Ever: 373 / Never: 214)
Total: 246 (Ever: 221 / Never: 25)
0
200
400
600
TMB (Nonsynonymous mutations / Mb)
EAS Never-smoker
EAS Ever-smoker
Western Never-smoker
Western Ever-smoker
median: 1.29
median: 2.39
median: 2.18
median: 5.96
***
ns
0.01
0.1
1
10
100
EAS
Never-smoker
Ever-smoker
Western
Never-smoker
Ever-smoker
B Somatic driver mutations
A panel of 15 core drivers
ALK
BRAF
EGFR
ERBB2
KEAP1
KRAS
MET
NF1
PIK3CA
PTEN
RB1
RBM10
SMARCA4
STK11
TP53
EAS never
EAS ever
West never
West ever
Mutated tumours (%), 95% CI
C Mutational signature
C>A transversions
C>A share of all substitutions
SBS1
SBS4
relative contribution
EAS Never-smoker
EAS Ever-smoker
Western Never-smoker
Western Ever-smoker
Ancestry comparison within never-smokers
Ancestry comparison within ever-smokers
EAS-enriched
ns
Western-enriched
Genome-wide comparison
-log10 adj. P
log2 odds ratio
EGFR
XIRP2
KEAP1
PTA1
RYR2
CSMD3
ADGRG4
ERICH3
LRP1B
CNTNAP4
KRAS
ADGRV1
SCN3A
RP1L1
CDH7
MUC16
SSPO
CAM
TTN
GOLGA6L2
RP11-1220K2.2
MUC12
ANKRD30B
MUC19
RGPD4
WIZ
GOLGA6L10
ZNF716
TRIM49B
ZNF729
D Cohort-level consistency
TMB
C>A
SBS1
SBS4
Chen2020
Xu2020
TCGA2014
TRACERx2023
Change from never to ever smoking (median, 95% CI)
Cross-ancestry
Within-EAS
Within-Western
EGFR
KEAP1
KRAS
TP53
ns
adj P < 0.05
Xu / TRACERx
Xu / TCGA
TCGA / TRACERx
Chen / Xu
Chen / TRACERx
Chen / TCGA
Ever
Never
log2 odds ratio